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Faculty of Environment and Technology

Musical Time Travel

A Data-Driven Analysis of Sonic Evolution and Cross-Generational Recommendation

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Abstract

Contemporary music recommendation systems are commonly shaped by collaborative filtering and popularity signals. Although these approaches are effective for personalisation, they can make historical or less-exposed recordings harder to discover when recommendations depend on prior engagement. This project investigates whether a content-based approach can support cross-generational music discovery by matching contemporary input tracks with sonically similar tracks from a user-selected historical decade. The resulting artefact, *The Time Machine*, combines a 112,358-track static corpus spanning 1970–2024 with six shared audio descriptors: acousticness, danceability, energy, loudness, tempo and valence. A k-nearest neighbours (k-NN) model uses normalised feature vectors and Euclidean distance to rank candidate tracks, while cosine similarity and Intra-List Similarity (ILS) provide complementary evaluation measures. The system is implemented as an interactive Streamlit dashboard containing both recommendation and exploratory data analysis components. Evaluation combines ten structured algorithmic test queries with a 22-participant user study. The benchmark produced a mean Euclidean distance of 0.0894, compared with a reported random baseline of 0.6264, representing an average distance 84.9% lower. The mean cosine similarity was 0.9985, and ILS was 0.9975. Human evaluation produced a mean perceptual similarity rating of 3.38/5 and a System Usability Scale score of 74/100. The findings provide encouraging evidence that simple, interpretable audio-feature similarity can support cross-generational discovery, while also demonstrating important limitations around cultural context, dataset coverage, and the incompleteness of six-dimensional audio representations. The project provides a transparent prototype, acknowledging that mathematical similarity is not equivalent to musical meaning.

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Chapter 1: Introduction

1.1 Context and Rationale

Music is a cultural record, a source of entertainment and a mechanism through which people retrieve autobiographical memories. The scale of contemporary streaming makes recommendation systems increasingly important in determining which parts of that musical record become visible to listeners. Spotify reported approximately 777 million monthly active users in the second quarter of 2026 (Spotify, 2026), illustrating the scale of automated discovery.

Collaborative filtering (CF) is a major recommendation paradigm. It learns from patterns of user interaction such as listening or rating behaviours. This is powerful because behaviour provides an implicit signal of preference, but it can also reinforce existing exposure patterns. Celma (2010) describes the long-tail problem in digital music recommendation, while Nguyen et al. (2014) found that the use of recommender systems can affect content diversity. CF also faces a cold-start problem when a track has insufficient interaction history (Typke et al., 2005). This is relevant to historical discovery because recordings released before the streaming era may have less comparable engagement information than contemporary releases.

This project explores content-based filtering (CBF) as a method to identify tracks from a chosen historical decade that share similar audio features with a given track. CBF represents an item through characteristics of the item itself. Here, the representation consists of six descriptors: acousticness, danceability, energy, loudness, tempo and valence. A historical track can therefore be compared without requiring a historical listening profile. This does not make CBF universally superior to CF; it makes it appropriate for testing whether sonic similarity can support cross-generational discovery.

The project also has psychological motivation. Janata (2009) investigated music-evoked autobiographical memory, while Pearce et al. (2013) describe music cognition in relation to mental time travel. Sedikides and Leunissen (2021) discuss the psychological benefits associated with music-evoked nostalgia. Jakubowski et al. (2020) found that music encountered during adolescence is disproportionately represented in autobiographical musical memories, while Krumhansl and Zupnick (2013) identified cascading effects involving older generations.

These findings motivate historical exploration but do not imply that an algorithm can reproduce nostalgia. Nostalgia is autobiographical and contextual. The narrower data-science question is whether an interpretable audio-similarity mechanism can provide historical candidates that listeners perceive as meaningfully related to a contemporary input.

1.2 Problem Statement and Scope

The problem addressed is the difficulty of deliberately discovering music from a chosen historical period based on sonic similarity. Engagement-driven recommendation can predict likely preferences, but it is not primarily designed to answer the question: “Which recording from a particular past decade sounds like this track?”

The project develops *The Time Machine*, a content-based prototype that accepts a track and a target decade, then returns the three nearest candidate tracks based on six normalised audio features. It is deliberately scoped as a research artefact rather than a production recommendation platform. It does not model long-term personal preference or claim to replace human judgement about musical meaning.

The working corpus contains 112,358 tracks. The first Kaggle dataset was restricted to 1970–2019 and the second to 2020–2024 (Kaggle, 2024a; Kaggle, 2024b). The 1970 lower boundary reflects the limited representation of earlier music in the available corpus and should be understood as a dataset limitation rather than a statement about the suitability of pre-1970 music for computational analysis.

The six features were selected because they are consistently available across both datasets. This improves reproducibility and avoids different representations between historical and contemporary records. The trade-off is that high-level descriptors cannot fully capture timbre, instrumentation, lyrics, culture, genre or personal meaning.

1.3 Aim, Objectives and Research Questions

The aim is to design, implement, and evaluate a content-based music recommendation system that supports psychologically motivated cross-generational discovery.

RQ1 — Algorithmic Efficacy: Can the system identify historical tracks that are perceived as sonically similar to modern inputs?

RQ2 — Usability and Performance: Is the interface intuitive and sufficiently responsive for interactive exploration?

RQ3 — Contextual Relevance: Do mathematically generated recommendations make sense to human listeners?

Five SMART objectives operationalise the aim:

- O1. — Data Pipeline: Construct a standardised 112,358-track corpus from the two Kaggle datasets, aligned on six shared audio features, by Week 3.
- O2. — Exploratory Analysis: Produce decade-level visualisations of audio-feature change across 1970–2024, including the Loudness War, by Week 4.
- O3. — Recommendation Engine: Implement decade-constrained k-NN using Euclidean distance for ranking, with cosine similarity and ILS as complementary measures, by Week 6.
- O4. — Artefact: Deploy the Streamlit dashboard with a sub-three-second response target, by Week 9.
- O5. — Evaluation: Evaluate the system using ten structured benchmark queries and a user study, by Week 10. The planned target was ten participants; 22 were obtained.

1.4 Report Structure

Chapter 2 critically reviews relevant literature. Chapter 3 translates the findings into system architecture and design decisions. Chapter 4 explains the implementation. Chapter 5 analyses quantitative and human-evaluation results against the research questions. Chapter 6 consolidates the findings, limitations and future work.

Chapter 2: Literature Review

2.1 Music Information Retrieval and Recommendation

Music Information Retrieval (MIR) includes computational methods for analysing and retrieving music. Recommendation is a single application, but its behavior depends on what is considered evidence of similarity. Collaborative filtering uses relationships between users and items, whereas content-based filtering uses item characteristics (Lops et al., 2011).

CF has practical advantages because interaction data can capture preferences that raw audio cannot, including artist loyalty and social trends. However, popularity and feedback effects can narrow exposure. Celma (2010) discusses the long-tail problem, while Nguyen et al. (2014) demonstrate that the use of recommender systems can affect content diversity. For historical discovery, the cold-start problem identified by Typke et al. (2005) is also relevant, because older recordings may have less comparable engagement data.

CBF provides a complementary solution: a candidate can be compared using its intrinsic characteristics without requiring an interaction history. A 1978 recording and a 2023 recording can therefore be represented in the same feature space. The limitation is equally important: if the representation omits lyrics, cultural context, or timbral detail, the model cannot recover those dimensions.

CBF is therefore selected as a problem-specific approach rather than presented as a universal replacement for CF. A hybrid model may ultimately provide stronger personalisation, but CBF offers the transparency needed to test whether audio similarity alone can support the targeted discovery task.

An important implication is that the definition of similarity determines the kind of discovery the system can produce. Behavioural similarity may recommend tracks that different users consumed in similar contexts, whereas audio similarity may recommend tracks that share measurable sonic properties despite having very different audiences. Neither definition is inherently “true”. For this reason, the present system is best interpreted as an alternative discovery lens. Its value lies partly in exposing relationships that a popularity or behaviour-driven system might not prioritise.

2.2 Audio Feature Representation and Validation

Spotify-derived high-level descriptors provide an accessible representation for computational music analysis. Schedl et al. (2014) describe the broader use of audio descriptors in MIR, while Panda et al. (2021) found that Spotify features such as valence and energy have useful relationships with broad affective dimensions but are less expressive than detailed signal-level representations.

The six project features capture complementary properties: acousticness, danceability, energy, loudness, tempo and valence. Together they form a compact description of a track's high-level sonic profile. However, Eerola and Vuoskoski (2011) show that musical emotion itself can be represented through different models, reinforcing the point that a small numerical vector cannot encode the full meaning of a recording. Aucouturier and Pachet (2002) likewise emphasise that music-similarity measures must be interpreted according to their intended use.

Consequently, a small Euclidean distance indicates that tracks are close in the selected six-dimensional representation; it does not mean that listeners will necessarily consider them interchangeable. Human evaluation is therefore used alongside geometric metrics.

Feature weighting is another unresolved issue. Min-max scaling gives each dimension a comparable numerical range, but it does not demonstrate that tempo should contribute exactly as much to perceived similarity as valence or acousticness. A future experiment could learn or estimate feature weights from participant judgements. Such an extension would turn the current equal-weight assumption into a testable modelling question and could improve alignment between algorithmic and perceptual similarity.

2.3 Sonic Evolution of Popular Music

Mauch et al. (2015) analysed more than 17,000 US pop recordings from 1960 to 2010 and demonstrated systematic changes in musical characteristics over time. This provides methodological precedent for treating music as a measurable time series.

The project applies a related approach to the available Spotify-derived features. Decade-level aggregation allows the dashboard to show changes in variables such as energy, loudness and tempo. Devine (2013) documents the Loudness War and the role of dynamic-range compression and mastering practices, providing theoretical grounding for the corresponding visualisation.

The EDA is exploratory rather than a causal test. A trend in the corpus does not, by itself, prove why it occurred, because genre composition, digitisation, and dataset coverage can also influence aggregate patterns.

2.4 Psychology and Neuroscience of Musical Time Travel

Janata (2009) investigated music-evoked autobiographical memory, while Pearce et al. (2013) describe music cognition as mental time travel. Sedikides and Leunissen (2021) discuss positive psychological effects associated with music-evoked nostalgia. Jakubowski et al. (2020) found disproportionate representation of adolescence-associated music in autobiographical memories, and Krumhansl and Zupnick (2013) identified cascading reminiscence effects involving older generations. Kudaravalli et al. (2024) relate these effects to neurocognitive and social development during adolescence.

These findings justify historical exploration, but not a universal assumption that a particular decade will be nostalgic for every user. Age, culture, and personal experience vary. The decade is therefore a user-controlled discovery context rather than a psychological prediction.

2.5 Algorithmic Bias and Ethical Dimensions

Recommendation systems can reproduce biases present in their source data. Ferraro et al. (2021) examine imbalance in music recommenders. The project's Spotify-derived datasets are shaped by platform availability, digitisation and metadata practices and should not be treated as a neutral representation of global musical history.

A second issue is contextual stripping. Six numerical descriptors can make culturally different recordings appear similar. This is not necessarily a computational error, but it becomes problematic if users interpret geometric similarity as cultural equivalence. Transparency about the representation is therefore part of responsible system design.

The ethical issue is therefore not solved by adding a disclaimer alone. Transparency informs the user, but it does not correct an unrepresentative corpus. A future version would need explicit coverage analysis and potentially rebalancing or stratified sampling to understand whose music is visible in the recommendation space. This distinction between disclosure and mitigation is important: acknowledging bias is necessary, but it is not equivalent to removing bias.

2.6 User Needs and HCI

Nielsen (1994) identifies response time as an important component of interactive usability, informing the project's three-second performance target. The distinction between measured and perceived latency is important: a user's experience of responsiveness is not identical to a server-side timing measurement. Brooke (1996) provides the standardised SUS instrument for assessing usability, and Nielsen and Landauer (1993) support small-sample usability testing, which informs the planned target of 10 participants. Ziegler et al. (2005) provide ILS as a list-level recommendation metric. The interface also includes a “Why This Match?” presentation to make the feature basis of recommendations visible.

2.7 Research Gap

The individual components used here—content-based recommendation, audio descriptors, k-NN, reminiscence research and usability evaluation—are established. The project's gap is their integration into a transparent, deployable artefact specifically designed for decade-targeted cross-generational discovery. The contribution is therefore an applied synthesis and evaluation rather than a claim to have invented a new recommendation algorithm.

Chapter 3: System Design and Artefact

3.1 Artefact Overview

The Time Machine is an interactive Streamlit dashboard with two functions. The recommendation component accepts a track and a target decade, then returns the three nearest candidates based on six normalised audio features. The Sonic Evolution component provides decade-level visualisations for the 1970–2024 corpus.

The architecture separates data preparation, recommendation logic and interface responsibilities. The core modules are data_pipeline.py, model.py and app.py. A separate evaluation_metrics.py script runs structured benchmark queries and is not required for normal user interaction.

3.2 Architecture and Module Responsibilities

data_pipeline.py loads the source CSVs, applies year boundaries, selects shared features, removes invalid or duplicate records, and exports the master dataset. model.py loads the dataset, normalises the six features and exposes get_recommendations(), which performs decade-constrained Euclidean k-NN and calculates complementary metrics. app.py manages user interaction, result presentation and EDA visualisations with a fourth standalone evaluation script (*evaluation_metrics.py*) used exclusively for dissertation benchmarking, not as part of the user-facing artefact. The separation allows the recommendation logic to be tested independently of the dashboard.

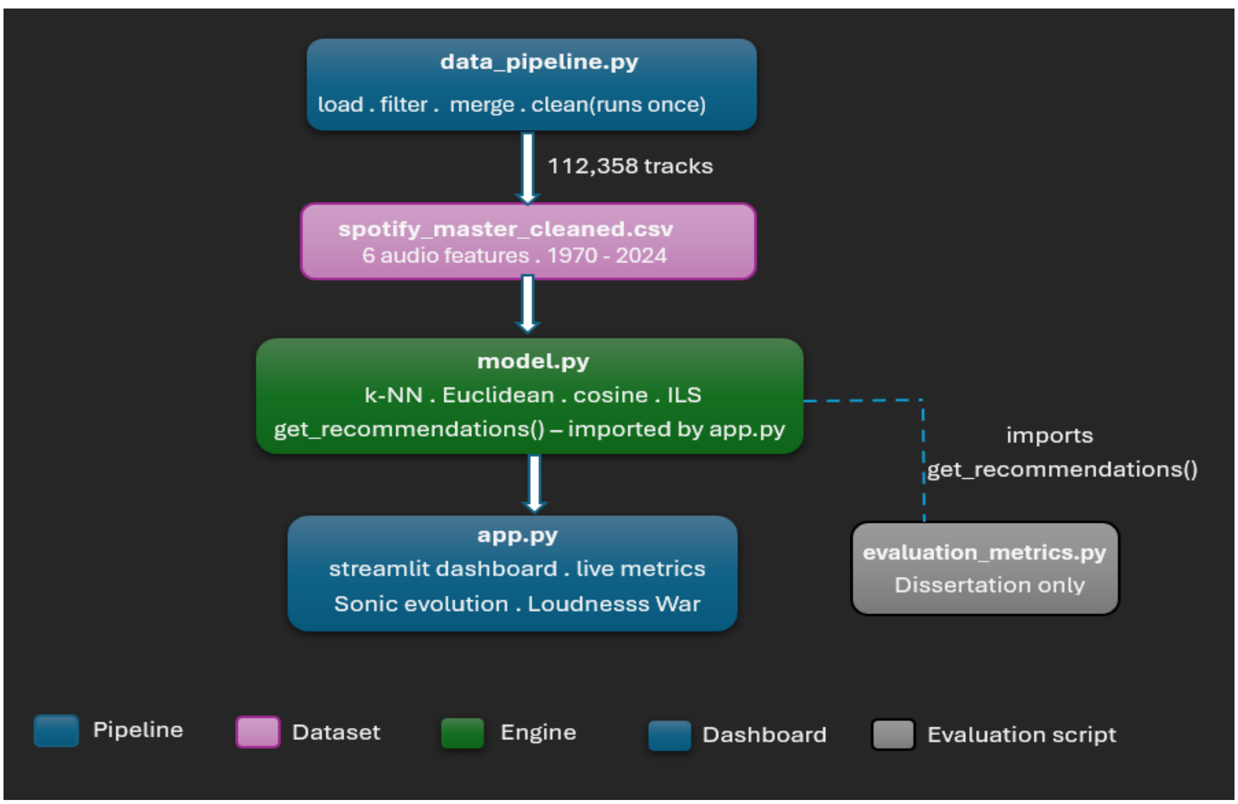

Figure 3.1: System architecture. The three core artefact modules are connected by solid lines; the evaluation script is a separate branch.

Module	Role and Responsibility
data_pipeline.py	Data ingestion and preparation. Loads both Kaggle CSV files, applies year filters (DS1: 1970–2019; DS2: 2020–2024), standardises the six shared audio features, removes duplicates and invalid values, and exports the master cleaned dataset (112,358 tracks).
model.py	The recommendation engine. Loads the master dataset, applies MinMaxScaler normalisation to the six features, and exposes the get_recommendations() function that performs k-NN search by Euclidean distance and computes cosine similarity and ILS on every query.
evaluation_metrics.py	Standalone evaluation script. Imports get_recommendations() from model.py and runs structured test cases across multiple input tracks and target decades, printing a full metric report for each query.
app.py	The user-facing Streamlit dashboard. Imports the recommendation engine, renders the search interface, displays results with per-result distance and cosine similarity scores, and presents the Sonic Evolution EDA charts and the Loudness War visualisation.

Table 3.1: Module responsibilities within The Time Machine system architecture.

3.3 Design Rationale

CBF was selected because it supports comparison of historical candidates without requiring historical engagement data. k-NN was chosen because it is non-parametric, interpretable, and appropriate for 112,358 records represented in only six dimensions. Euclidean distance is used for ranking after normalisation; cosine similarity provides a complementary directional measure, and ILS assesses the coherence of the recommendation list.

A richer MIR model could capture timbre and other detail, but would make the system more complex and less transparent.

The original design included live API retrieval, but the project switched to static datasets after relevant changes to the Spotify API. The static approach removes network dependency from the recommendation path and allows caching, but it also makes corpus coverage fixed.

3.4 Dashboard Design

The interface follows three principles: simple input, transparent output and contextual exploration. Users select a track and decade without needing to understand the algorithm. Results include feature-level information, while the Sonic Evolution component provides wider context. The design deliberately presents similarity as a property of the chosen representation rather than as an objective statement about musical meaning.

Chapter 4: Implementation

4.1 Data Preparation and Corpus Construction

Dataset 1 was restricted to 1970–2019 and Dataset 2 to 2020–2024 (Kaggle, 2024a; Kaggle, 2024b). The ranges provide a consistent temporal boundary for the merged corpus. The datasets were aligned on the six shared audio features and relevant metadata. Latin-1 encoding was used to preserve non-ASCII artist information. Invalid or unusable feature records were removed, and duplicates were eliminated, resulting in a final corpus of 112,358 tracks.

The transformation is implemented in data_pipeline.py rather than performed inside the dashboard. This improves reproducibility and reduces the risk that the interface and evaluation script operate on different data versions.

4.2 Feature Normalisation

Distance-based algorithms are sensitive to scale. Tempo is measured in beats per minute and loudness on a decibel scale, while several other features are bounded between zero and one. Without scaling, the numerical magnitude of a feature could cause it to dominate the Euclidean distance.

Min-max scaling was therefore applied:

x' = (x - xmin) / (xmax - xmin)

The scaler was fitted on the complete corpus rather than separately by decade. This ensures that the same raw feature value receives the same transformed value regardless of the target era. Scaling does not make the features perceptually equal; it only makes their numerical ranges comparable. Each selected dimension consequently contributes to distance on the same [0,1] scale.

Using a single scaler across the entire corpus also raises a methodological consideration. The transformation depends on the observed minimum and maximum values in the combined dataset. If future data contain values outside those bounds, the scaling procedure would need to be refitted or handled explicitly. For this prototype, fitting once on the fixed corpus is appropriate because reproducibility of the experimental representation is more important than continuous production updates.

4.3 End-to-End Recommendation Procedure

The query begins by identifying the requested track and extracting its six normalised features as a vector. The candidate database is then filtered to the selected decade. The model therefore solves a constrained nearest-neighbor problem rather than finding a globally nearest track and filtering afterward.

For each candidate y, Euclidean distance is calculated as:

d(x,y) = sqrt(sum from i=1 to 6 of (xi - yi)^2)

Candidates are ranked by distance from smallest to largest; self-matching is excluded where necessary, and the three nearest tracks are returned. Cosine similarity is calculated as:

cos(theta) = (x · y) / (||x|| ||y||)

ILS is also calculated to provide a list-level perspective. The three measures are complementary: Euclidean distance measures point-to-point separation, cosine measures vector direction, and ILS considers coherence within the recommendation set.

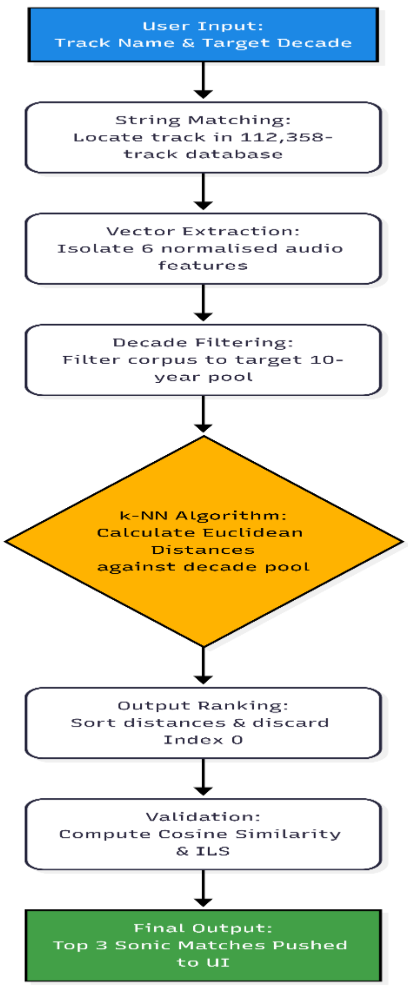

Figure 4.1: End-to-end query pipeline from user input to recommendation output.

When the user selects "Travel Through Time", the following background sequence executes:

Step 1 — Track lookup: The system performs a case-insensitive string match to locate the user's requested song in the database. Once found, the algorithm discards the track's metadata and retains only its six normalised audio features. Conceptually, the input track is now a single coordinate in a six-dimensional feature space.

Step 2 — Decade filtering: The system filters the 112,358-track corpus to retain only tracks released within the target decade (usually leaving a "decade pool" of 11,000 to 23,000 tracks).

Step 3 — k-NN ranking: Four neighbours are requested (n_neighbors=4) because the source may reside in the decade pool; index 0 is discarded. The three remaining candidates are ranked in ascending order of Euclidean distance. **Euclidean distance is the sole ranking criterion.**

Step 4 — Validation metrics: Cosine similarity measures proportional feature alignment independently of absolute magnitude; ILS (Ziegler *et al.*, 2005) measures pairwise coherence across the three results. A random baseline — three randomly sampled decade tracks — provides the denominator for the percentage-below-random metric.

Step 5 — Why This Match?: Per-feature similarity percentages are computed as (1 - abs(src_val - rec_val)) * 100, converting normalised differences to an interpretable similarity percentage for each of the six features.

4.4 Evaluation Design

Evaluation combines quantitative and human evidence because no single measure can establish successful music recommendation. Ten structured queries across different input tracks and target decades were benchmarked using Euclidean distance, a reported random reference, percentage improvement, cosine similarity, and ILS.

The random comparison is descriptive rather than inferential because the available benchmark does not report repeated random draws, confidence intervals, or significance testing. The 84.9% mean reduction is therefore interpreted as a strong result for the selected queries, not population-level statistical proof.

A 22-participant study involved five trials per participant and collected perceptual similarity ratings, SUS responses, and relevance/speed feedback. The study was intended to determine whether geometric similarity transferred to human judgement and whether the interface was usable.

4.5 Dashboard Implementation and Performance

Streamlit caching reduces the need for repeated dataset loading and preprocessing. The application avoids live network calls during recommendation, removing external API latency from the critical path. The project target was a response time below three seconds, consistent with Nielsen's (1994) interactive usability threshold. Sussman and Sekuler (2022) demonstrate that even moderate delays measurably impair executive function, reinforcing why responsiveness matters beyond convenience. Participants also rated perceived speed. Because a complete instrumented latency log was not recorded during development, perceived responsiveness is reported separately from formally timed computational measurement.

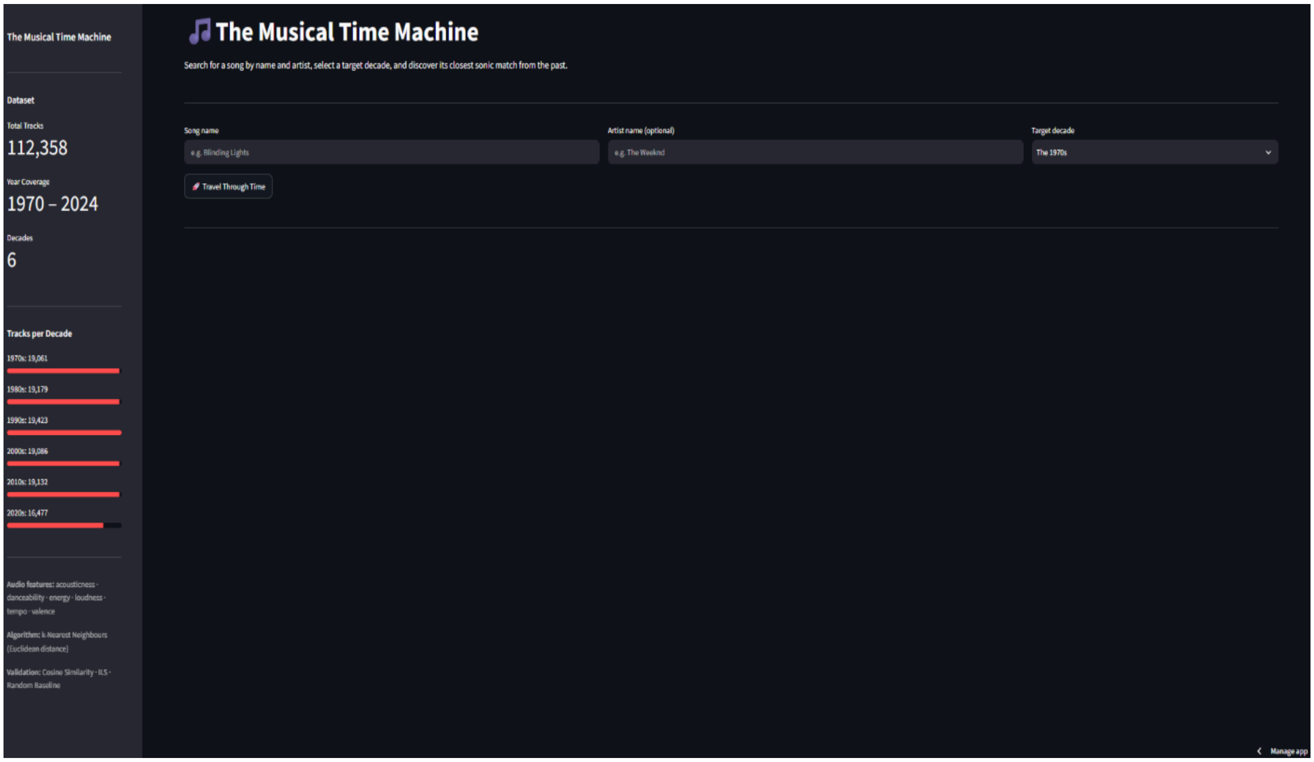

Figure 4.2: The Time Machine Streamlit dashboard on initial load, showing the sidebar dataset overview, search inputs, and decade selector.

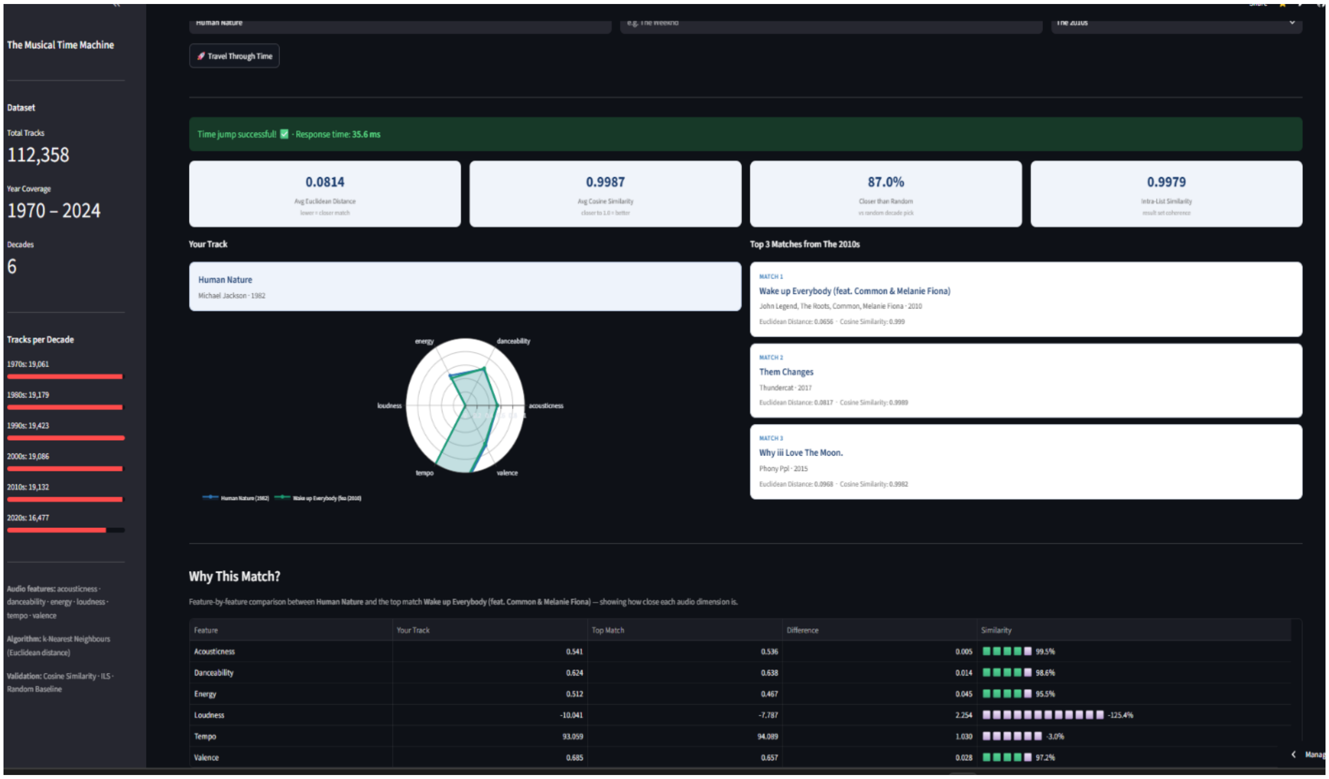

Figure 4.3: Dashboard following a successful query, showing metric cards, radar chart, and result cards.

Figure 4.4: Sonic Evolution section showing decade trend lines with searched track highlighted.

4.6 Ethical Considerations and Limitations

The project uses public datasets and anonymised participant evaluation data collected under the UWE Bristol ethical approval process. The principal ethical issue is representational bias: the source datasets do not constitute a balanced sample of global music. A second limitation is contextual stripping because six variables cannot encode lyrics, language, social significance or personal associations.

A further limitation is item coverage. One participant could not find a requested track because it was absent from the static corpus. Thus, the CBF approach reduces dependence on historical engagement data but does not solve the separate problem of missing items. The system can only recommend tracks represented in the corpus and can only process input tracks with available feature information.

4.7 Tools and Libraries

The implementation uses Python 3.12, pandas, scikit-learn, Streamlit and Plotly. Separating the data, model and interface components supports reproducibility and allows the recommendation logic to be evaluated independently.

Chapter 5: Evaluation

5.1 Quantitative Benchmark

Ten structured queries tested the recommendation engine across different contemporary tracks and historical target decades. The mean Euclidean distance of the selected recommendations was 0.0894, compared with a reported mean random-baseline distance of 0.6264. This corresponds to an average 84.9% reduction in distance. Individual improvements ranged from 72.9% for Levitating targeted to the 1970s to 93.3% for Bad Guy targeted to the 1990s.

The result demonstrates that the implementation consistently identifies candidates close to the input according to the selected six-feature representation. It does not, however, establish that listeners will necessarily perceive those candidates as strongly similar. The human evaluation is therefore necessary.

The mean cosine similarity of 0.9985 and ILS of 0.9975 indicate high directional and list-level similarity in the normalised feature space. These values should not be interpreted as percentages of musical similarity. High cosine values can arise naturally when non-negative feature vectors occupy similar directions. They are best treated as complementary measures of consistency.

The lower improvement for Levitating → 1970s is plausibly related to differences between the contemporary electronic-pop profile and the candidate population, although the ten-query benchmark is too small to establish a causal explanation. Overall, the benchmark supports the technical claim that the k-NN implementation reliably finds close feature-space neighbours within the selected decade.

A further limitation is that the benchmark is query-based rather than a full corpus-level retrieval evaluation. Metrics such as precision@k, recall@k or nDCG would require a defined relevance ground truth for historical music, which is not available in the current project. Human relevance judgements could provide such a ground truth in future work, allowing the system to be evaluated using standard recommender-system ranking measures as well as distance-based metrics.

Input Track	Target	Euclidean	Random	% Below	Cosine	ILS
Blinding Lights	1980s	0.0858	0.7295	88.2%	0.9986	0.9990
Levitating	1970s	0.1281	0.4732	72.9%	0.9972	0.9967
Shape of You	1990s	0.1012	0.5882	82.8%	0.9988	0.9986
Bohemian Rhapsody	2010s	0.0854	0.4603	81.4%	0.9976	0.9938
Stay With Me	2000s	0.1001	0.7403	86.5%	0.9983	0.9975

Rolling in the Deep	1980s	0.0807	0.6885	88.3%	0.9991	0.9982
Watermelon Sugar	1970s	0.0886	0.4636	80.9%	0.9989	0.9985
Bad Guy	1990s	0.0480	0.7132	93.3%	0.9997	0.9994
Someone Like You	2000s	0.0796	0.6529	87.8%	0.9981	0.9960
Hotline Bling	1980s	0.0963	0.7545	87.2%	0.9983	0.9971
Mean	—	0.0894	0.6264	84.9%	0.9985	0.9975

Table 5.1: Quantitative results across ten test queries. Euclidean distances computed in normalised six-dimensional space.

5.2 RQ1: Algorithmic Efficacy and Perceptual Similarity

Participants rated the three recommendations on a five-point perceptual-similarity scale. Match #1 received a mean of 3.46 (SD = 1.15), Match #2 3.38 (SD = 1.27), and Match #3 3.31 (SD = 1.20), giving an overall mean of 3.38.

The ordering is encouraging because the closest algorithmic result received the highest mean rating. However, the differences are small and the standard deviations indicate substantial participant variation. Without inferential testing or a correlation between distance and individual ratings, the findings should not be described as statistically validated. They provide preliminary evidence that the model's ranking broadly corresponds with human perception.

Participant comments identified tempo, rhythm, energy and acoustic range as important similarity cues. These observations are broadly consistent with the model's feature set because tempo and energy are directly represented. At the same time, listeners may use vocals, lyrics, instrumentation or cultural knowledge, none of which is fully represented. RQ1 is therefore cautiously supported: the system can produce historically positioned tracks that listeners perceive as moderately similar, but the six-dimensional Euclidean distance is not a complete model of human musical similarity.

Match Position	Mean (1–5)	Std. Dev.	% Rating ≥ 3
Match #1 (closest)	3.46	1.15	69%
Match #2	3.38	1.27	62%
Match #3 (furthest)	3.31	1.20	61%
Overall mean	3.38	—	64%

Table 5.2: Mean Likert perceptual similarity ratings per match position (n=22).

"They share same beat and almost same baseline rhythm. They carry similar energy." — Participant 5 (2026)

"Felt the most similar because they shared a similar acoustic range to the chosen song." — Participant 4 (2026)

5.3 RQ2: Usability and Performance

The System Usability Scale score was 74.0/100, above the commonly cited 68-point benchmark (Brooke, 1996). Item-level results were particularly positive for ease of use (4.23), integration (4.31) and learning quickly (4.31). Confidence was lower at 3.92, suggesting that the interface could do more to explain the meaning of the results.

System speed received the highest performance rating of 4.46/5, and perceived latency was reported as below 1 second in the modal response. These results are consistent with the static-data architecture and caching strategy. However, perceived latency is not equivalent to a formally instrumented server-response measure, so the conclusion is limited to perceived responsiveness. Future testing should record repeated timings under defined hardware and deployment conditions.

RQ2 is therefore supported with respect to perceived usability and responsiveness. The system did not create a significant interaction barrier for the study participants, but further usability work is warranted.

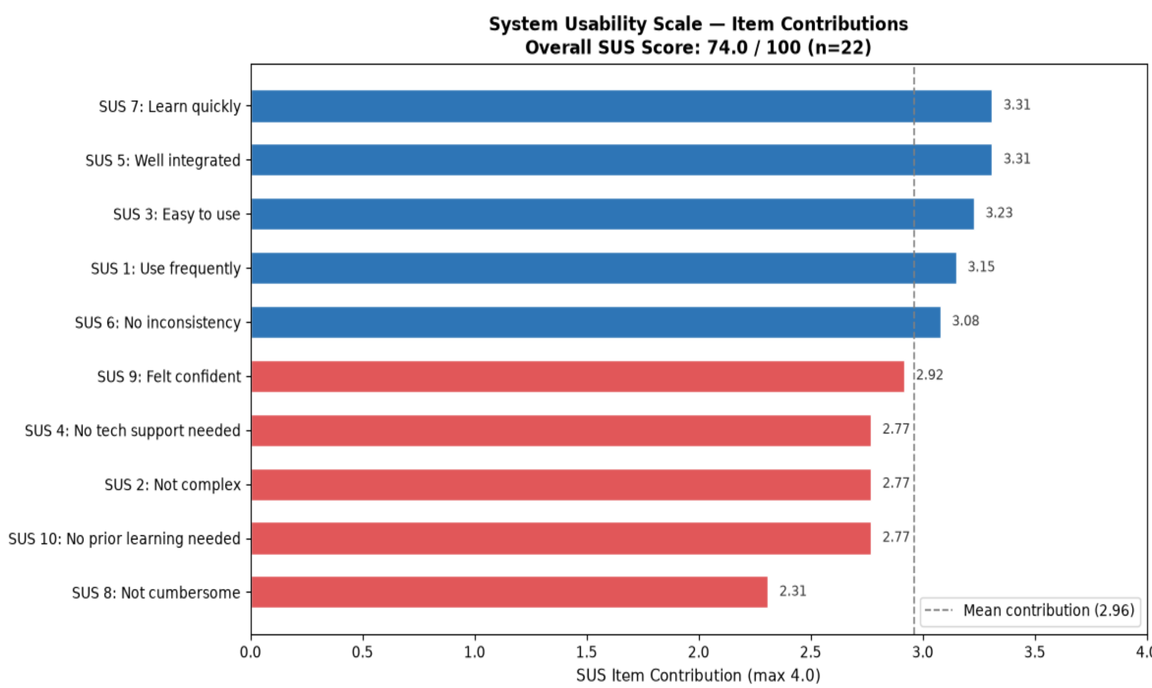

Figure 5.1: System Usability Scale item contributions (n=22). Overall SUS score = 74.0, exceeding the above-average usability threshold of 68 (Brooke, 1996).

Dimension	Mean / Modal	Interpretation
System speed (Q22)	4.46 / 5	Highest-rated dimension in the questionnaire
Overall relevance (Q23)	4.15 / 5	Majority agreed recommendations logically made sense
Dashboard trends (Q24)	3.77 / 5	Moderate agreement — visualisations useful but not universally intuitive
Perceived latency (Q25)	Modal: < 1 second	Well within 3.0-second threshold (Nielsen, 1994)

Table 5.3: Mean responses to Part 3 system performance questions (n=22). Q25 uses a latency bracket scale, not a Likert agreement scale.

5.4 RQ3: Contextual Relevance

The mean relevance rating was 4.15/5, indicating that participants generally considered the recommendations logically related to their input tracks. This is important because relevance is broader than mathematical proximity.

The evaluation also demonstrated the limits of sonic-only matching. An example involving Shape of You and a 1990s Latin-pop recommendation shows how similar tempo, energy or danceability can produce a mathematically strong result that a listener may find culturally unexpected. This illustrates the difference between sonic and contextual similarity.

The Sonic Evolution visualisations received a lower mean score of 3.77/5. One participant described the visualisation as useful but “a little complex”. This suggests that the recommendation path was easier to understand than the exploratory analytics layer. Clearer annotation, fewer simultaneous dimensions and progressive disclosure would improve this component.

Another participant reported that the requested song could not be found. This exposes the static corpus boundary: reducing dependence on interaction histories does not solve missing-item coverage. RQ3 is therefore supported overall, but with a clear requirement for better contextual representation and broader data coverage.

5.5 Integrated Discussion

Taken together, the evidence indicates that *The Time Machine* is technically functional and useful as an exploratory prototype. The benchmark demonstrates consistent nearest-neighbour behaviour, the human ratings show moderate perceived sonic commonality, SUS indicates good usability, and the relevance score indicates that users generally understood the recommendations.

The principal contribution is not state-of-the-art recommendation accuracy. It is a transparent demonstration that established MIR techniques can support a specific discovery task without requiring a historical engagement profile. Recommendations can be traced to a defined representation and distance calculation.

The evaluation also demonstrates why the quality of recommendations cannot be reduced to a single metric. The 84.9% reduction in reported distance is strong for the selected queries but is limited by the benchmark size and baseline methodology. The very high cosine values do not capture cultural relevance. The perceptual mean of 3.38 is encouraging but also shows that mathematical proximity does not guarantee strong human agreement. SUS of 74 indicates good usability rather than perfection.

The project therefore succeeds at three different levels to different degrees: algorithmically, it performs consistent nearest-neighbour search; perceptually, users generally perceive some similarity; contextually, recommendations are often relevant but can fail when culture and semantics are absent from the representation. The principal limitations are dataset coverage, feature simplicity, corpus bias and evaluation scale.

Chapter 6: Conclusion

6.1 Conclusion

This project investigated whether content-based audio similarity could support cross-generational music discovery through a transparent and psychologically motivated artefact. *The Time Machine* integrates a 112,358-track corpus spanning 1970–2024, six high-level audio features, a normalised k-NN engine and a Streamlit interface with exploratory visualisations.

Across ten structured queries, recommendations had a mean Euclidean distance of 0.0894, compared with a reported random baseline of 0.6264, corresponding to an 84.9% reduction. Mean cosine similarity was 0.9985 and ILS was 0.9975. These results demonstrate consistent behaviour within the selected feature space.

Human evaluation produced mean perceptual-similarity ratings of 3.46, 3.38 and 3.31 for the three ranked positions, with an overall mean of 3.38/5. The trend is consistent with the model ranking but does not justify a claim of statistical validation. Usability was stronger: SUS was 74/100, system speed was rated 4.46/5 and overall relevance 4.15/5.

RQ1 is therefore supported cautiously: the system can generate historical candidates that listeners perceive as moderately similar. RQ2 is supported by the perceived usability and responsiveness. RQ3 is also supported overall, although culturally unexpected matches demonstrate that sonic similarity cannot fully capture the context of music.

The five objectives were substantively achieved: the corpus was constructed, EDA implemented, the k-NN engine deployed, the dashboard developed and evaluation completed with 22 rather than the planned ten participants. The findings nevertheless define clear boundaries around what the system can claim.

Domain	Metric	Result	Target
Quantitative	Mean % below random baseline	84.9%	> 0% (outperform chance)
Quantitative	Mean Cosine Similarity	0.9985	Closer to 1.0 = better
Quantitative	Mean ILS	0.9975	Closer to 1.0 = better
Qualitative	Mean Likert Rating	3.38 / 5	Above midpoint (3.0)
Qualitative	SUS Score	74.0 / 100	Above 68 (Brooke, 1996)
Qualitative	System Speed Rating	4.46 / 5	Highest-rated dimension
System Perf.	Modal Perceived Latency	< 1 second	< 3.0 seconds (Nielsen, 1994)

Table 6.1: Consolidated evaluation results.

6.2 Contribution and Limitations

The main contribution is an integrated and explainable prototype rather than a new algorithm. The project demonstrates how established MIR concepts can be combined with a psychologically motivated discovery task and evaluated at geometric, perceptual, usability and contextual levels.

The main limitations are representation, data coverage and evaluation scale. Six features cannot encode all dimensions of musical meaning. The public datasets are not culturally comprehensive, and the static corpus prevents automatic access to missing or new tracks. Ten benchmark queries and 22 participants provide useful prototype evidence but cannot establish broad generalisation across listeners, genres and cultures.

6.3 Future Work

Future work should test richer MIR representations, such as MFCCs and other spectral descriptors, and compare them against the current six-feature model using human-rated evaluation. A hybrid CBF/CF model could preserve historical discovery while incorporating individual preference. A broader and more culturally diverse corpus could reduce representational bias.

Evaluation should also be expanded through repeated random baselines, confidence intervals, inferential tests and systematic sampling across decades and genres. Correlation between algorithmic distance and human ratings would provide stronger evidence for RQ1. Finally, the EDA interface could be simplified through progressive disclosure, clearer annotation and feature explanations.

6.4 Closing Reflection

The project demonstrates that engagement-driven recommendation is not the only way to navigate musical history. A content-based representation provides a transparent route into the past. However, the findings also show that numerical similarity is only one dimension of musical meaning. *The Time Machine* is therefore best understood as a discovery aid: it provides mathematically grounded starting points while leaving cultural interpretation, memory and personal judgement with the listener.

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Appendices

Appendix A: Records of Meetings

This section contains the records of the meetings held with my supervisor. All meetings were conducted online via Microsoft Teams.

Date	Discussion	Outcome / Action
02/04/2026	First supervisory meeting. Project expectations, scope and meeting format discussed.	Meeting frequency and 30-minute session length agreed. Project direction confirmed.
26/05/2026	Proposal finalisation, ethical approval, participant information sheet and increase in user-study participant target discussed.	Participant target increased from 5 to 10. Ethical approval form submitted. Consent forms finalised.
05/06/2026	Spotify API deprecation and its impact. Recent MIR literature and post-proposal priorities.	Static Kaggle datasets adopted. Newer journal sources identified to strengthen literature review.
24/06/2026	Dashboard feedback. Bar chart duplication and dashboard connectivity raised.	Duplicate bar chart removed. Sonic Evolution updated to highlight searched track on decade trend lines.
20/07/2026	Dashboard review: radar chart, evaluation metrics, colour scheme, Loudness War boxplot and questionnaire design.	Radar chart updated to overlay source and top match. Cosine similarity and ILS clarified as validation metrics. Boxplot added. Latency question refined.
24/08/2026	Chapter 4 written feedback: excessive standard detail, missing end-to-end model explanation, ethics structure, citation formatting, word count.	Chapter 4 restructured: end-to-end walkthrough added; ethics split into three subsections; standard library explanations removed; citations corrected to UWE Harvard format.
25/08/2026	Final pre-submission review. Table placement, bibliography formatting, appendix structure.	Tables moved inline. Bibliography DOI: and [Accessed dates] corrected. Meetings appendix reformatted. Report submitted.

Appendix B: Ethical Approval

Section1: Students Details	
Student Record ID	14211
Faculty	College of Arts, Technology and Environment
Department	
Supervisor First Name	Kat
Supervisor Last Name	Fair
I confirm that I have completed the Research Data Management Training	Yes
Project Details Proposed Project Title	Musical Time Travel: A Data-Driven Analysis of Sonic Evolution and Cross-Generational Recommendation
Level of Study	PGT
Faculty	College of Arts, Technology and Environment
Department	
Module Code	UFCF9Y-60-M
Module Name	CSCT Masters
Module Leader	Rachel Long

Student Name	David Adebayo (Student)
Student Number	25001142
Project Details – Record Questions A.	
Are human participants involved?	Yes
Have you seen and approved the UWE participant information sheet?	Yes
Have you seen and approved the UWE consent form?	Yes
Have you seen and approved the research instrument (e.g.questionnaire/interview schedule)?	Yes
B. Does the proposed research fall into any of the following categories?	
Research involving potentially vulnerable groups:(*except for educational research in classroom settings where safeguarding arrangements are already in place, and where appropriate parent and pupil consent for the proposed UWE research activity is also in place), (*N.B. Applicable to UWE Education Department students only), people with a learning disability or cognitive impairment, those who lack decision making capacity or people in a dependent or unequal relationship?	No
Research involving NHS or independent hospital patients, social care (including care/nursing homes) service users or prisoners?	No
Research involving deception or which is conducted without participants' full and informed consent at the time the study is carried out, for example studies using data from social media	No
Research involving the collection of data, or access to data/ records,involving personal or sensitive confidential information, including genetic or other biological information concerning identifiable individual or Special Category Data under the Data Protection Act, or linkage of datasets with the result that individuals can be identified	No
Research which would or might induce psychological stress, anxiety or humiliation, or cause more than minimal pain or distress to either participants or researchers	No
Research involving intrusive interventions or data collection methods,e.g. the administration of substances, taking of biological samples, vigorous physical exercise or techniques such as hypnotism. This includes where participants are persuaded to reveal information they would not otherwise disclose in the course of their everyday lives or within public forums	No
Research involving visual/vocal methods, where participants or other individuals may be identifiable in the visual images used or generated	No
Research which may involve data sharing of confidential information beyond the initial consent given, e.g. where the research topic or data gathering involve a risk of information being disclosed that would require researchers to breach confidentiality conditions agreed with participants	No
Research on, or which may elicit information about, sensitive topics,including but not limited to sex and sexuality, security sensitive information including extreme beliefs or views, experiences of abuse or harm, religious beliefs, drug use, criminal activity, ethnicity/racism	No
Special Categories	
Research involving human tissue, including body parts blood, and cells	No
Research using administrative data not in the public domain, secure data or security-sensitive data	No
Research where data collection is undertaken or shared outside the UK,or where personal data will cross National boundaries	No
Research involving animals (including invertebrates) or animal by-products	No
Research which might have a negative environmental impact	No
Research involving politically and/or culturally sensitive funding sources or partners	No
Research involving financial inducements (other than reasonable expenses and compensation for time)	No